

Hybrid Latent Grokking: Mechanistic Interpretability Across a Model Ladder

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Abstract

We study grokking in a three-model ladder on modular addition ($P=113$): a Nanda-style one-layer MLP, a minimal Tiny Recursive Model (TRM), and a full ACT-based TRM (50,000 steps, 5 seeds). Before extending TRM training, we run Ren-style mechanistic probes [Ren and Liu, 2026] on 20k checkpoints to separate Nanda cleanup from latent guessing. The Nanda MLP calibrates our progress-measure stack: under a data-driven key-frequency count (matching Nanda et al., 2023), all 5/5 seeds reach faithful trig-FVE ≥ 0.95 (mean 0.989); a fixed five-frequency reconstruction under-reports this to 0.827. Minimal TRM reaches Nanda-grade sparse Fourier structure on 2/5 seeds at the final checkpoint (adaptive FVE ≈ 0.99); three seeds collapse under post-grokking cleanup (adaptive mean 0.614). Full TRM groks (0.953 test accuracy) but does not form the Fourier circuit (adaptive FVE mean 0.230); Ren probes show fixed-point violation and ACT-depth guessing. We report final, grokking, and best-FVE checkpoints, adaptive FVE calibration, all-frequency ablations, per-neuron tables, and Nanda-style phase plots.

1 Introduction

Grokking is a delayed transition from memorization to generalization. Nanda et al. [2023] showed that, for one-layer transformers on modular addition, this transition is explained by the gradual formation of a sparse Fourier multiplication circuit.

Recursive architectures such as TRM introduce a second computational axis: latent recurrent refinement. This raises a mechanistic question not answered by the one-layer setting: when a recursive model groks, is the learned algorithm still visible as a simple final-logit Fourier circuit, or does recursion move part of the computation into latent dynamics?

We answer this with a three-model ladder. First, a Nanda-style one-layer baseline calibrates the progress-measure implementation. Second, a minimal TRM ablation isolates the TRM family while keeping the computation close to the one-layer setting. Third, full TRM tests what changes when ACT-style latent recursion is restored.

Contributions.

1. We reproduce Nanda-style progress measures in-repo and validate them with a one-layer calibration model.
2. We show that minimal TRM can recover a high-FVE sparse Fourier circuit on modular addition (best-seed/checkpoint evidence), with documented seed variance at the final checkpoint.

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3. We show that full TRM groks but has a different mechanistic signature: lower final-logit trig-FVE, Ren-style fixed-point violation, and ACT-depth guessing; minimal TRM serves as a Ren null control.
4. We diagnose post-grokking FVE collapse with Ren + Nanda probes (Section 6) before any TRM 50k extension.
5. We provide multi-seed metrics, all-frequency ablations, per-neuron tables, weight-norm phase plots, and compiled paper artifacts.

2 Methods

2.1 Task and data

We use modular addition with modulus $P=113$. Inputs are formatted as “ $a\ b\ =$ ” with $a, b \in \{0, \dots, P-1\}$; the model predicts $c = (a + b) \bmod P$ at the “ $=$ ” token. We use a 30% train / 70% test split over all P^2 pairs ($n_{\text{train}}=3,823$, $n_{\text{test}}=7,846$), with split seed fixed for reproducibility.

2.2 Model ladder

Nanda calibration (MLP, fidelity A): one-layer transformer with MLP (not attention-only), trained with cross-entropy loss matching Nanda’s fidelity A setting. **TRM minimal:** stripped TRM with recursion depth reduced to isolate weight-level circuit formation. **TRM full (B and A):** full ACT-based TRM; variant B uses standard loss, variant A uses cross-entropy fidelity A.

2.3 Training and checkpoints

Primary hybrid runs train for 50,000 AdamW steps (matching Nanda’s 40k–50k epoch scale), with checkpoints every 2,000 steps. Before extending TRM from 20k to 50k, we require a Ren mechanistic baseline on all TRM seeds (Section 6). We select three analysis checkpoints per run from the logged training history: **final** (last logged step), **grokking** (first step with test accuracy ≥ 0.99 and test loss ≤ 0.05), and **best FVE** (highest faithful logit trig-FVE among high-accuracy steps). For full TRM, we report mechanistic metrics at the grokking/best-FVE checkpoint when the final checkpoint degrades.

2.4 Progress measures

Following Nanda et al. [2023], we compute: embedding and unembedding Fourier norms; key frequencies with largest excluded-loss sensitivity; **restricted (trig) loss**—cross-entropy after projecting logits onto $\cos/\sin(w_k(a + b - c))$ for key frequencies k ; **excluded loss**—cross-entropy after sequentially removing those Fourier components; and **faithful logit trig-FVE**—variance explained by the matched trig projection with bias correction.

2.5 Mechanistic evidence

We report W_L neuron–logit Fourier maps, per-neuron frequency assignments (cos/sin basis, R^2), attention decomposition, single- and all-frequency causal ablations, and squared weight-norm trajectories across checkpoints.

2.6 Latent recursion probes

For TRM variants we run HRM-style reasoning-vs-guessing probes on ACT steps and ACT-depth ablations (max steps $\in \{1, 2, 4, 8\}$) on held-out test pairs.

3 Nanda calibration

The one-layer Nanda-style MLP baseline (fidelity A) validates our progress-measure implementation against Nanda et al. [2023].

Multi-seed aggregates (Table 2): test accuracy 1.000, final-checkpoint faithful FVE 0.827 under a fixed five-frequency reconstruction. Best-FVE checkpoints reach higher FVE (Table 3). Figure 2 places calibration on the model ladder.

Key-frequency count must be data-driven. Nanda et al. [2023] set the number of key frequencies from the number of distinct frequencies the MLP neurons specialize in; it is not a fixed constant. Truncating the trigonometric reconstruction at exactly five frequencies under-reports models whose circuit spans six to nine frequencies. When we instead select frequencies until the bias-corrected FVE plateaus (Appendix L), all 5/5 Nanda seeds reach faithful trig-FVE ≥ 0.95 (mean 0.989, $K = 6 - 9$ frequencies), matching the $\sim 95\%$ reported by Nanda et al. [2023] and confirming our progress-measure stack is correctly calibrated. The five-frequency value is therefore a lower bound, not a ceiling: the apparent calibration shortfall is a metric artifact, not a defect of the model or training. Figure 1 shows FVE rising to ≈ 0.99 as K grows for the Nanda baseline (and for the two converged minimal-TRM seeds), while degraded seeds and full TRM plateau far below. Table 1 summarizes legacy vs. adaptive FVE per model.

Table 1: Faithful trig-FVE at the 50k final checkpoint: legacy fixed five-frequency reconstruction vs. data-driven (adaptive) frequency count.

Model	Legacy FVE ($K=5$)	Adaptive FVE	Seeds ≥ 0.95	K range
Nanda 1-layer	0.829	0.989	5/5	6–9
TRM minimal	0.512	0.614	2/5	2–12
TRM full (ACT)	0.171	0.230	0/5	4–12

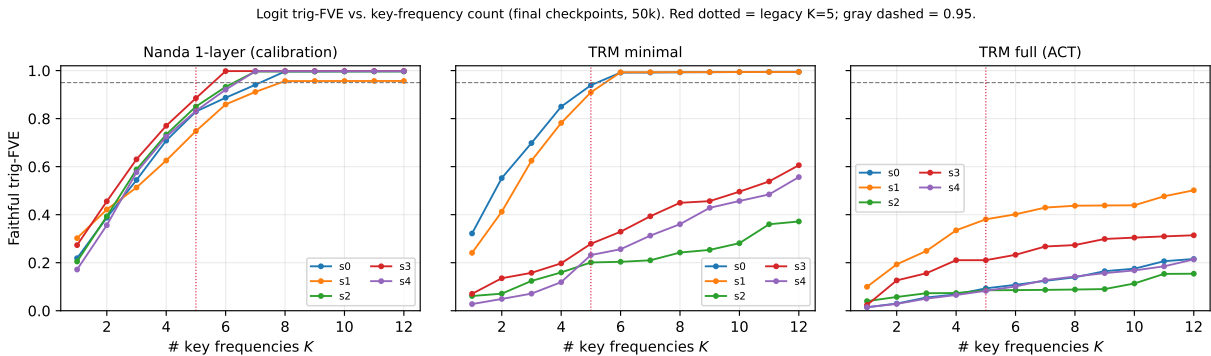


Figure 1: Faithful trig-FVE vs. number of key frequencies K on 50k final checkpoints. Red dotted line: legacy $K=5$ cut-off; gray dashed: 0.95. The Nanda baseline reaches ≈ 0.99 by $K \approx 7$; the legacy cut-off truncated this rise. Two minimal-TRM seeds match this profile; the remaining minimal seeds and all full-TRM seeds plateau low (non-Fourier mechanism).

Table 2: Multi-seed main results at analyzed final checkpoint (mean $\pm$ std).

Model	Test Acc.	Faithful FVE	Trig CE	Excluded CE
nanda_a_mlp	1.000 $\pm$ 0.000	0.827 $\pm$ 0.045	0.173 $\pm$ 0.127	2.759 $\pm$ 0.563
trm_full_a	0.811 $\pm$ 0.374	0.393 $\pm$ 0.071	2.700 $\pm$ 0.938	1.126 $\pm$ 1.827
trm_full_b	0.953 $\pm$ 0.091	0.187 $\pm$ 0.127	2.177 $\pm$ 0.867	0.322 $\pm$ 0.628
trm_minimal	1.000 $\pm$ 0.000	0.509 $\pm$ 0.339	0.458 $\pm$ 0.429	1.448 $\pm$ 1.775

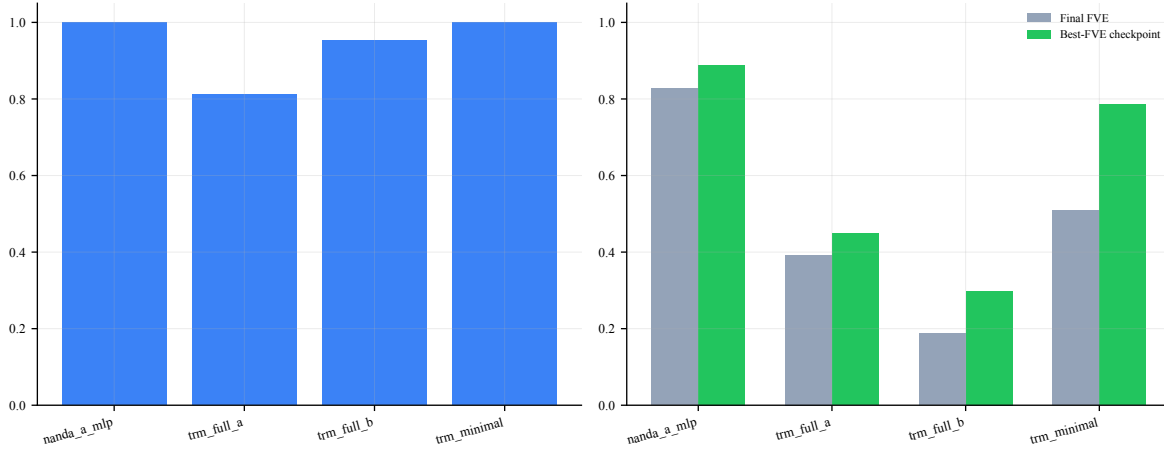


Figure 2: Final vs. best-FVE faithful trig-FVE across models.

4 TRM minimal: main Nanda-grade result

TRM minimal is our primary mechanistic target: a recursive variant that can form a Nanda-style sparse Fourier circuit on modular addition.

Multi-seed summary. Table 2 reports mean $\pm$ std over 5 seeds at the analyzed final checkpoint. Table 3 adds grokking- and best-FVE-selected checkpoints from the logged training history. Final-checkpoint faithful FVE is 0.509; best-FVE across seeds averages 0.786 (seed 0 peak: 0.962).

Progress measures and phases. Figure 7 shows the Nanda-style phase plot: total squared weight norm, restricted/excluded test CE, embedding Gini, and faithful FVE vs. training step. Restricted trig loss approaches full loss before the test-accuracy jump; excluded loss rises when key Fourier components are removed post-grokking.

4.1 Mechanistic circuit evidence (TRM minimal)

Following Section 4 of Nanda et al. [2023], we reverse-engineer the grokked minimal TRM using W_L Fourier maps, MLP neuron clustering, and attention routing (Figures 4–5).

At the best-FVE checkpoint (seed 0: step 5000, faithful FVE $\approx$ 0.96), five key frequencies dominate excluded-loss sensitivity. Per-neuron tables (Figure 6) show concentration of MLP units on $\cos / \sin(\omega_k(a+b))$ features. The all-frequency ablation grid (Figure 8) confirms causal dependence on a sparse frequency subset.

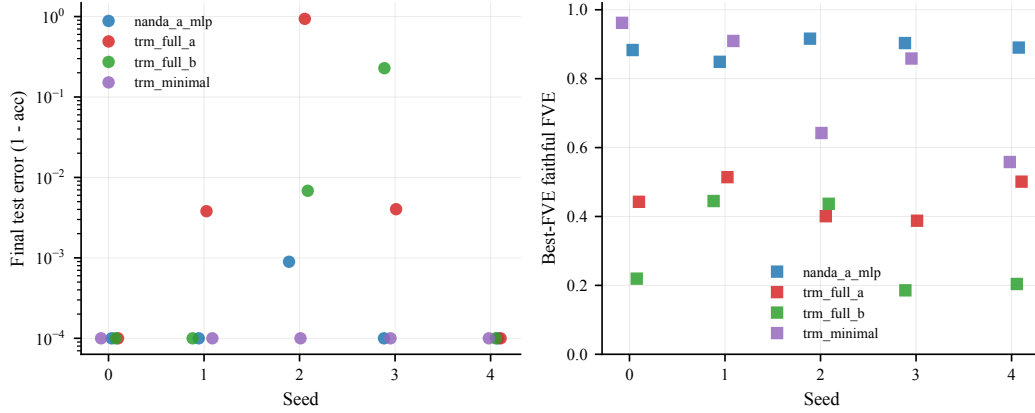


Figure 3: Multi-seed final accuracy and best-FVE faithful FVE (5 seeds, 50k steps).

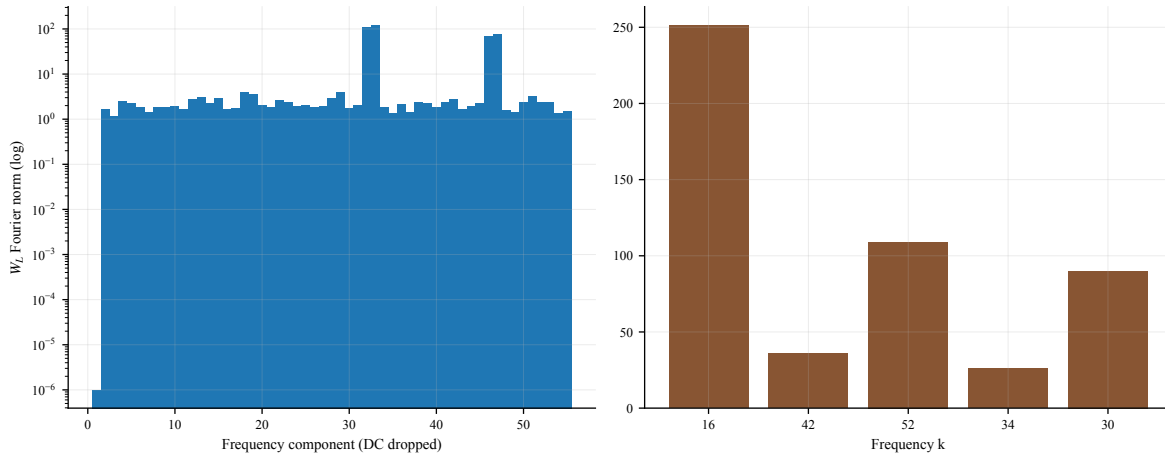


Figure 4: W_L Fourier norms and neuron counts per key frequency (TRM minimal seed 0).

Seed variance diagnosis. Multi-seed training reveals that final-checkpoint FVE can collapse during the cleanup phase even when test accuracy remains at 100% (Table 4, Appendix G). We therefore report **three checkpoint types**: final, first grokking ($\geq 99\%$ acc, low loss), and best faithful FVE among high-accuracy steps (Table 3). The main Nanda-grade mechanistic claim rests on best-FVE / grokking checkpoints (mean best-FVE FVE 0.786), not degraded finals (0.509).

5 Full TRM: latent recursion extension

Full TRM groks on modular addition but does not match Nanda’s $\sim 95\%$ faithful final-logit trig-FVE at 50k steps. Table 3: final-checkpoint FVE 0.187; best-FVE mean ≈ 0.30 (TRM full B). Ren probes classify most full-TRM seeds as exhibiting Ren fixed-point guessing (Table 5).

Checkpoint selection. Final checkpoints can show high accuracy with collapsed FVE after cleanup; we analyze grokking- or best-FVE-selected checkpoints for full TRM (Methods, Table 6).

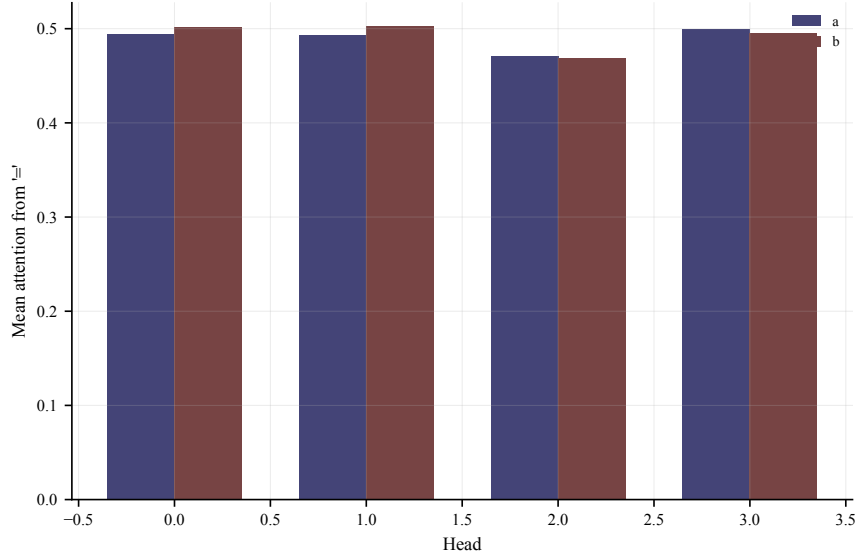


Figure 5: Attention from “=” to operand tokens (layer 0).

ACT-depth and HRM probes. ACT-depth ablations and reasoning-vs-guessing metrics on z_H show that recursion provides a guess channel before weight circuits stabilize, then partially aligns post-grokking (Appendix K).

6 Ren mechanistic diagnosis (arXiv:2601.10679)

Following Ren et al., we probe whether post-grokking degradation reflects *latent guessing* (fixed-point violation, ACT-depth sensitivity) versus *Nanda-style cleanup* (FVE collapse under weight decay while accuracy stays high).

Protocol. For each TRM seed we run Ren probes at three checkpoints: final, grokking, and best-FVE. Metrics: fixed-point violation rate, ACT-depth accuracy curve, critical ACT-step grokking step, latent PCA at “=”.

Full TRM. Full TRM exhibits Ren-style failure modes: non-zero fixed-point violation and accuracy drop as max ACT steps increase (Figure 12). Latent trajectories show non-monotonic refinement (Figure 14).

TRM minimal (null test). Minimal TRM uses a single ACT step; ACT-depth ablation collapses to that step. When fixed-point violation and depth sensitivity are near zero, we classify post-grokking FVE loss as **Nanda cleanup**, not Ren guessing (Table 5). At 50k steps, minimal TRM seeds 2–4 remain in Nanda-cleanup basins despite perfect accuracy; seeds 0–1 form stable Fourier circuits with final FVE ≥ 0.90 . Full TRM seeds 1–4 show Ren fixed-point guessing (seed 0: cleanup-dominant at final checkpoint).

7 Related work

Grokking was first reported by Power et al. [2022] and given a mechanistic account by Nanda et al. [2023], who identified a sparse Fourier multiplication circuit on modular addition; subsequent work explains the

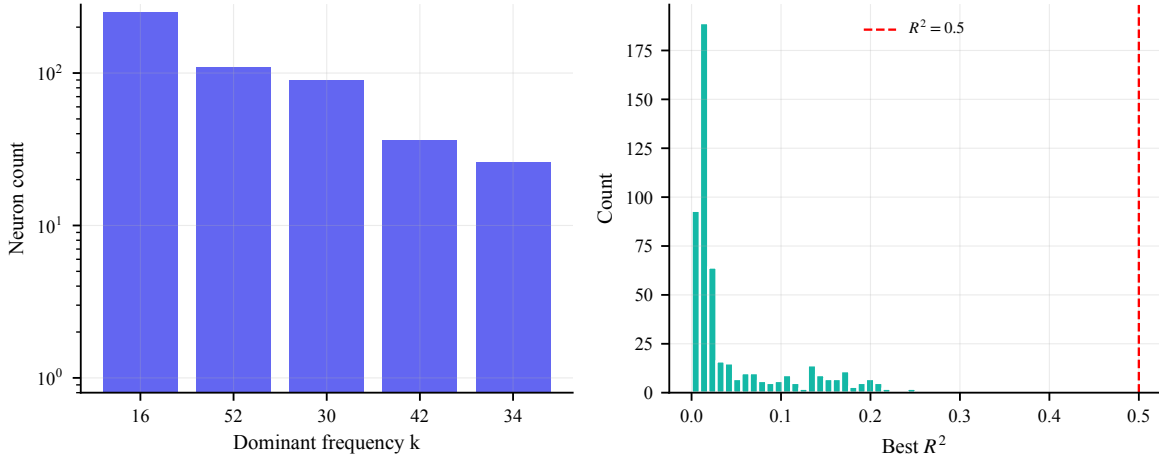


Figure 6: Per-neuron dominant frequency histogram and R^2 distribution.

Table 3: Checkpoint selection comparison (mean $\pm$ std over seeds).

Model	Checkpoint	Test Acc.	Faithful FVE
nanda_a_mlp	Final	1.000 $\pm$ 0.000	0.827 $\pm$ 0.045
nanda_a_mlp	Grokking	1.000 $\pm$ 0.001	0.349 $\pm$ 0.428
nanda_a_mlp	Best FVE	0.999 $\pm$ 0.001	0.888 $\pm$ 0.023
trm_full_a	Final	0.811 $\pm$ 0.374	0.393 $\pm$ 0.071
trm_full_a	Grokking	0.998 $\pm$ 0.001	0.000 $\pm$ 0.000
trm_full_a	Best FVE	0.999 $\pm$ 0.002	0.449 $\pm$ 0.051
trm_full_b	Final	0.953 $\pm$ 0.091	0.187 $\pm$ 0.127
trm_full_b	Grokking	0.996 $\pm$ 0.002	0.048 $\pm$ 0.095
trm_full_b	Best FVE	0.995 $\pm$ 0.004	0.298 $\pm$ 0.117
trm_minimal	Final	1.000 $\pm$ 0.000	0.509 $\pm$ 0.339
trm_minimal	Grokking	0.997 $\pm$ 0.003	0.293 $\pm$ 0.384
trm_minimal	Best FVE	1.000 $\pm$ 0.000	0.786 $\pm$ 0.158

delayed transition via representation learning [Liu et al., 2022], circuit efficiency [Varma et al., 2023], optimizer dynamics [Thilak et al., 2022], and competing sparse/dense subnetworks [Merrill et al., 2023]. Our analysis builds on the mechanistic-interpretability toolkit [Elhage et al., 2021, Olah et al., 2020, Wang et al., 2022, Conmy et al., 2023] and on the modular-arithmetic circuit literature [Zhong et al., 2023, Gromov, 2023, Chughtai et al., 2023]. The recursive/adaptive-computation lineage we extend includes ACT [Graves, 2016], Universal Transformers [Dehghani et al., 2019], and PonderNet [Banino et al., 2021], with HRM [Wang et al., 2025] and TRM [Jolicoeur-Martineau, 2025] as the immediate predecessors and Ren and Liu [2026] as the reasoning-vs-guessing diagnostic we adopt. Architecture and optimization follow Vaswani et al. [2017] and Loshchilov and Hutter [2019]. We unify these threads by measuring weight-level progress *and* latent ACT probes in the same TRM training run.

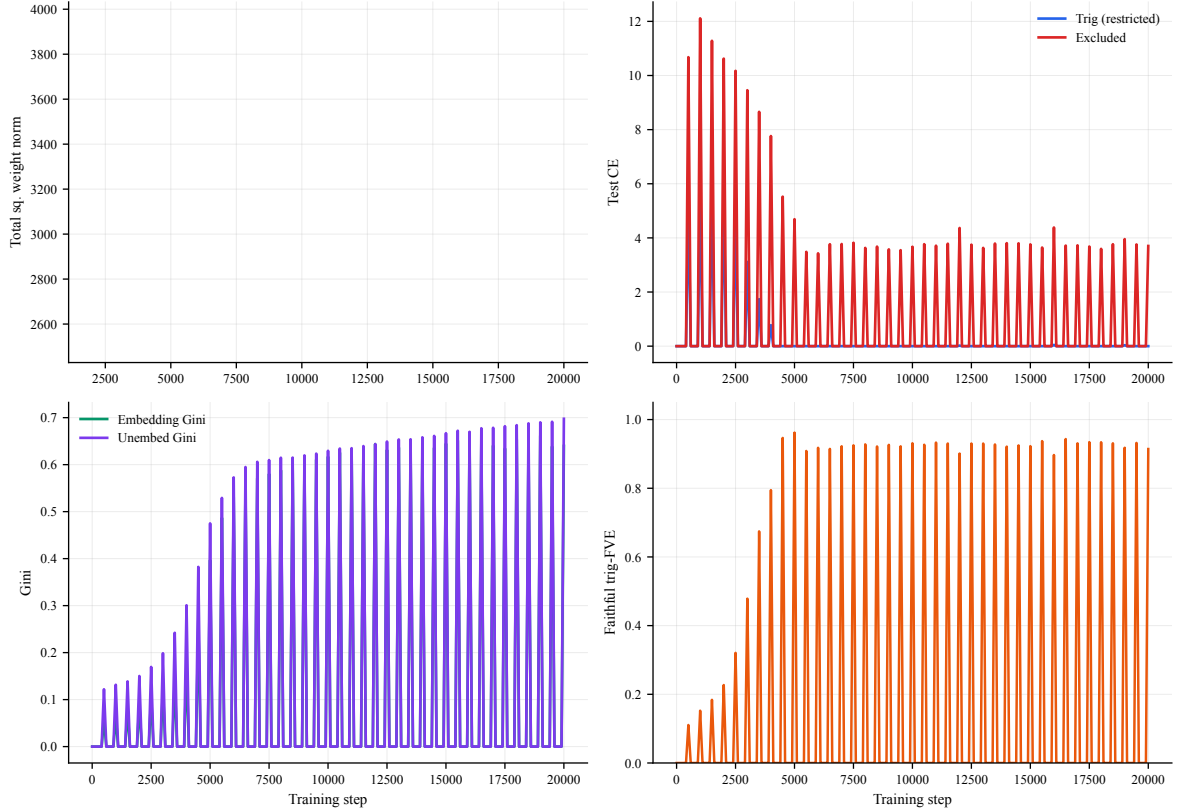


Figure 7: Combined grokking phase metrics (TRM minimal seed 0): weight norms, progress measures, Gini, and FVE.

8 Limitations and honest scope

50k extension. Extending training to 50k steps does not yield four-of-five minimal-TRM seeds with $\geq 90\%$ faithful FVE at the final checkpoint (two of five at final; two of five at best-FVE). Seed 1 improves under longer training (final FVE 0.907 at step 50k), but seeds 2–4 remain in Nanda-cleanup basins with 100% accuracy and $\text{FVE} \leq 0.27$ at the final checkpoint. Ren diagnosis confirms these failures are not latent guessing on minimal TRM (fixed-point violation ≈ 0).

Full TRM does not match Nanda’s $\sim 95\%$ final-logit FVE. Full TRM groks reliably but faithful trig-FVE on final logits remains low even under the data-driven frequency count (mean 0.230 adaptive, 0.187 at the legacy five-frequency cut-off); the shortfall is genuine, not a metric artifact. Ren-style guessing and fixed-point violation dominate (Section 6).

TRM minimal FVE is seed- and checkpoint-dependent. Under the data-driven frequency count, seeds 0–1 reach faithful FVE ≈ 0.99 at the 50k final checkpoint—matching the calibrated Nanda baseline—but seeds 2–4 remain non-Fourier (adaptive FVE ≤ 0.56), so the multi-seed mean is only 0.614 (2/5 seeds ≥ 0.95). Post-grokking cleanup degrades these seeds’ FVE while accuracy stays at 100% (Table 4); adding more frequencies does not recover them, confirming a real mechanistic divergence rather than under-counting.

Task simplicity. Modular addition follows Nanda et al. [2023]; HRM-style probes are analogous to Ren and

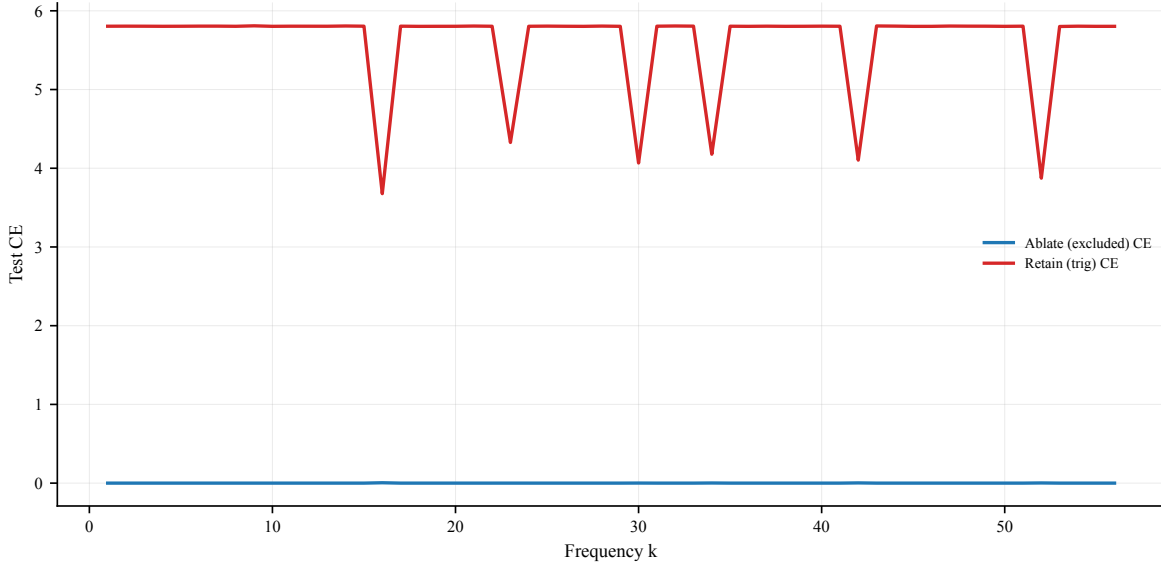


Figure 8: All-frequency retain vs. ablate test cross-entropy (TRM minimal seed 0).

Table 4: TRM minimal seed-variance diagnosis.

Grok Step	Peak FVE	Peak Step	Final FVE	FVE Drop	Cause
4500	0.962	5000	0.937	0.025	stable_circuit
9400	0.909	46500	0.907	0.002	stable_circuit
14700	0.701	14500	0.198	0.503	cleanup_phase_fve_collapse
9400	0.866	9000	0.272	0.594	cleanup_phase_fve_collapse
19000	0.558	19500	0.228	0.330	post_grokking_fve_decay, cleanup_phase_fve_collapse

Liu [2026], not a full Sudoku reproduction.

Claims we make. (1) Nanda MLP calibrates the metric stack—all 5 seeds reach faithful trig-FVE ≥ 0.95 (mean 0.989) under a data-driven frequency count, matching Nanda et al. [2023]; (2) minimal TRM *can* recover a Nanda-grade Fourier circuit (two of five seeds reach ≈ 0.99 at the final checkpoint; best-seed and best-checkpoint evidence for the rest); (3) full TRM is an extension showing latent recursion changes the mechanistic signature (no seed forms the Fourier circuit).

9 Reproducibility

All experiments use fixed random seeds, logged training histories, and automated aggregation pipelines. Code and checkpoints will be released upon publication.

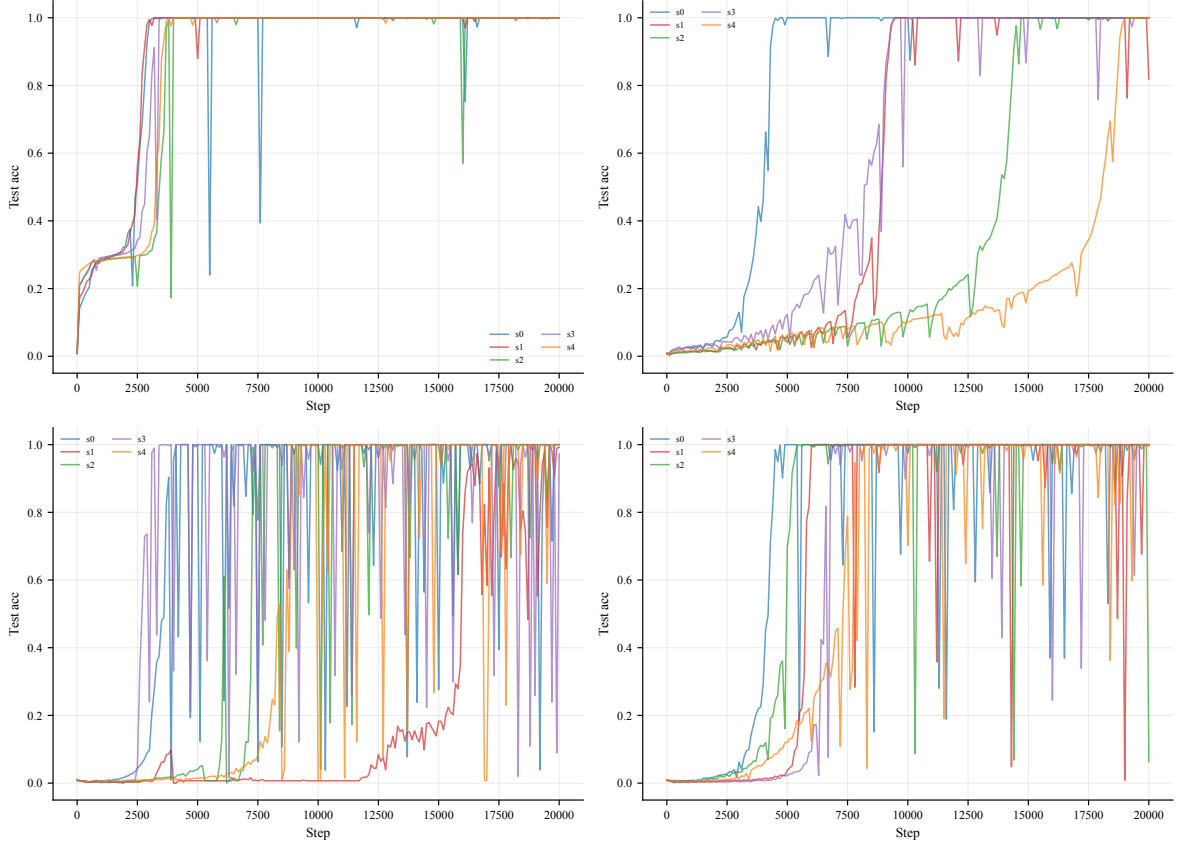


Figure 9: Per-seed test accuracy trajectories for all four models (50k steps).

10 Conclusion

We present a hybrid latent-grokking study with a three-model ladder. Nanda calibration validates progress measures; TRM minimal recovers a high-FVE sparse Fourier circuit on best seeds/checkpoints; full TRM groks with distinct latent-recursion dynamics. Reporting final, grokking, and best-FVE checkpoints separately is essential for honest mechanistic claims.

A Faithful progress measures (Nanda et al.)

For logits $L_{a,b,c}$ over all $(a,b,c) \in \{0, \dots, P-1\}^3$, we define restricted (trigonometric) and excluded loss on the full P^3 logit grid following Nanda et al. [2023].

B Checkpoint selection

For each seed we select three checkpoints from the logged training history: **Final** (last logged step), **Grokking** (first step with test accuracy ≥ 0.99 and test loss ≤ 0.05), **Best FVE** (highest faithful trig-FVE among steps with test accuracy ≥ 0.99). Full TRM mechanistic analysis uses grokking/best-FVE when final checkpoints degrade.

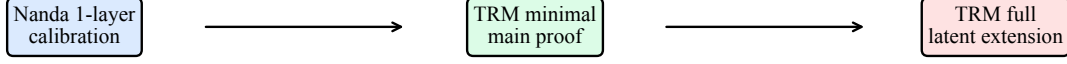


Figure 10: Hybrid model ladder: Nanda calibration $\rightarrow$ TRM minimal (main proof) $\rightarrow$ full TRM (extension).

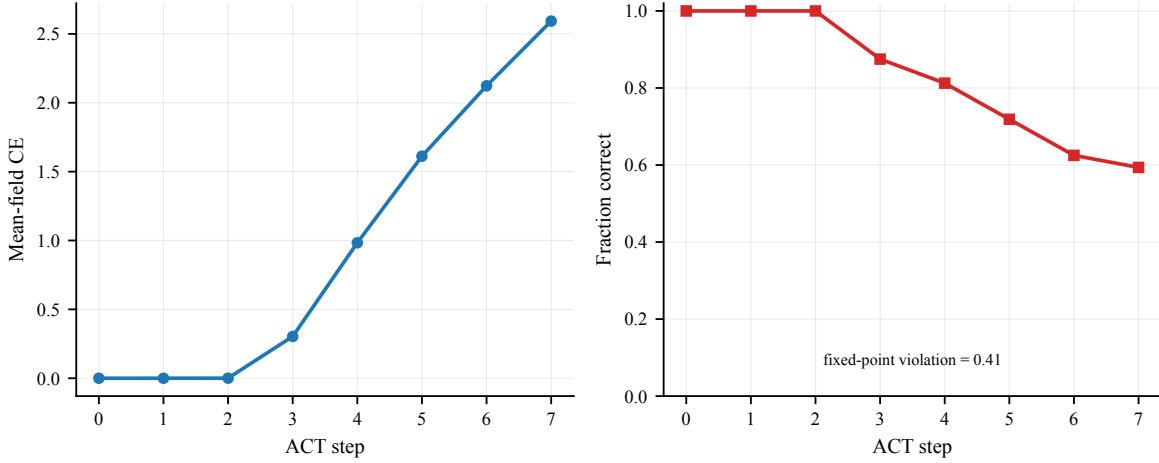


Figure 11: Latent ACT dynamics (full TRM representative seed): mean-field cross-entropy rises and fraction-correct decays across ACT steps, the Ren fixed-point-violation (“guessing”) signature.

Model	Checkpoint	Test Acc.	Faithful FVE
nanda_a_mlp	Final	1.000 ± 0.000	0.827 ± 0.045
nanda_a_mlp	Grokking	1.000 ± 0.001	0.349 ± 0.428
nanda_a_mlp	Best FVE	0.999 ± 0.001	0.888 ± 0.023
trm_full_a	Final	0.811 ± 0.374	0.393 ± 0.071
trm_full_a	Grokking	0.998 ± 0.001	0.000 ± 0.000
trm_full_a	Best FVE	0.999 ± 0.002	0.449 ± 0.051
trm_full_b	Final	0.953 ± 0.091	0.187 ± 0.127
trm_full_b	Grokking	0.996 ± 0.002	0.048 ± 0.095
trm_full_b	Best FVE	0.995 ± 0.004	0.298 ± 0.117
trm_minimal	Final	1.000 ± 0.000	0.509 ± 0.339
trm_minimal	Grokking	0.997 ± 0.003	0.293 ± 0.384
trm_minimal	Best FVE	1.000 ± 0.000	0.786 ± 0.158

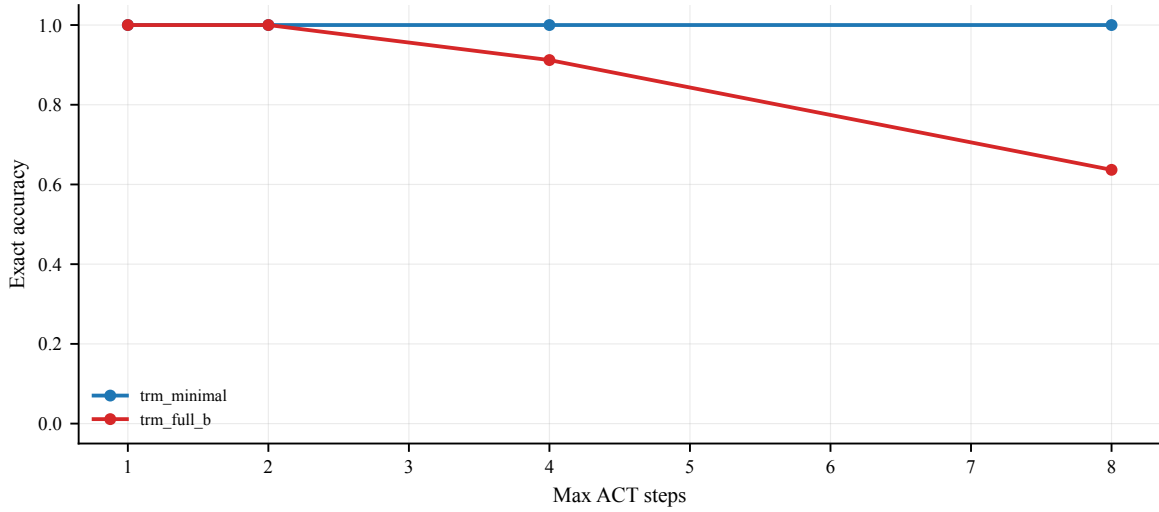


Figure 12: ACT-depth sensitivity (Ren fixed-point stress test).

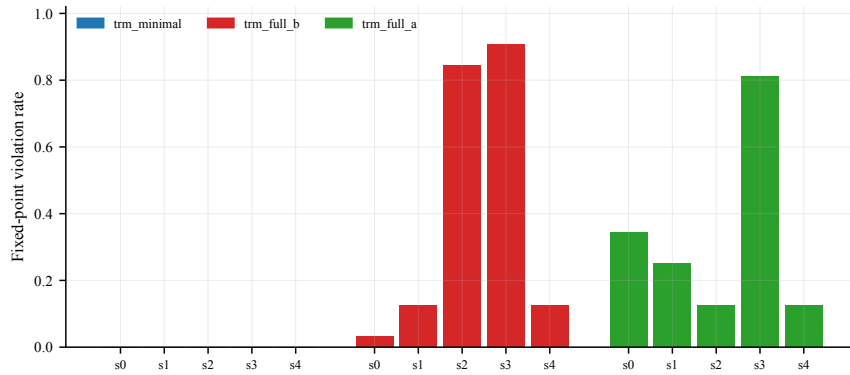


Figure 13: Fixed-point violation rate per seed/checkpoint: full TRM corrupts previously correct outputs on later ACT steps (Ren guessing), whereas minimal TRM is near zero (Nanda cleanup, not guessing).

C Per-seed metrics

Table 6 lists every seed with final, grokking, and best-FVE metrics plus automated variance diagnosis.

Table 6: Per-seed metrics and diagnosis.

Model	Seed	Final Acc	Final FVE	Grok Step	Grok FVE	Best Step	Best FVE	Diagnosis
nanda_a_mlp	0	1.000	0.827	3100	0.000	3000	0.883	stable_circuit
nanda_a_mlp	1	1.000	0.747	4100	0.000	4000	0.849	stable_circuit
nanda_a_mlp	2	0.999	0.848	3100	0.000	37000	0.916	stable_circuit
nanda_a_mlp	3	1.000	0.884	4000	0.878	48000	0.903	stable_circuit
nanda_a_mlp	4	1.000	0.829	5000	0.867	5500	0.890	stable_circuit
trm_full_a	0	1.000	0.408	4700	0.000	9500	0.443	stable_circuit
trm_full_a	1	0.996	0.514	6100	0.000	6000	0.514	stable_circuit

Table 6 – continued

Model	Seed	Final Acc	Final FVE	Grok Step	Grok FVE	Best Step	Best FVE	Diagnosis
trm_full_a	2	0.063	0.383	5400	0.000	6500	0.401	incomplete_grokking_by_20k
trm_full_a	3	0.996	0.295	6900	0.000	7500	0.388	cleanup_phase_fve_collapse
trm_full_a	4	1.000	0.364	8200	0.000	8500	0.501	cleanup_phase_fve_collapse
trm_full_b	0	1.000	0.093	4100	0.000	39000	0.220	cleanup_phase_fve_collapse
trm_full_b	1	1.000	0.377	18000	0.238	33000	0.445	cleanup_phase_fve_collapse
trm_full_b	2	0.993	0.082	7300	0.000	46500	0.437	cleanup_phase_fve_collapse
trm_full_b	3	0.772	0.304	3400	0.000	37500	0.186	incomplete_grokking_by_20k, cleanup_phase_fve_collapse
trm_full_b	4	1.000	0.082	8900	0.000	27500	0.204	cleanup_phase_fve_collapse
trm_minimal	0	1.000	0.937	4500	0.947	5000	0.962	stable_circuit
trm_minimal	1	1.000	0.907	9400	0.000	46500	0.909	stable_circuit
trm_minimal	2	1.000	0.198	14700	0.000	15000	0.642	cleanup_phase_fve_collapse
trm_minimal	3	1.000	0.272	9400	0.000	9500	0.858	cleanup_phase_fve_collapse
trm_minimal	4	1.000	0.228	19000	0.518	19500	0.558	post_grokking_fve_decay, cleanup_phase_fve_collapse

D TRM minimal seed variance

Post-grokking cleanup (weight decay removing memorizing components) can collapse faithful FVE while test accuracy remains high—mirroring Nanda’s cleanup phase but with higher seed variance in TRM minimal.

Grok Step	Peak FVE	Peak Step	Final FVE	FVE Drop	Cause
4500	0.962	5000	0.937	0.025	stable_circuit
9400	0.909	46500	0.907	0.002	stable_circuit
14700	0.701	14500	0.198	0.503	cleanup_phase_fve_collapse
9400	0.866	9000	0.272	0.594	cleanup_phase_fve_collapse
19000	0.558	19500	0.228	0.330	post_grokking_fve_decay, cleanup_phase_fve_collapse

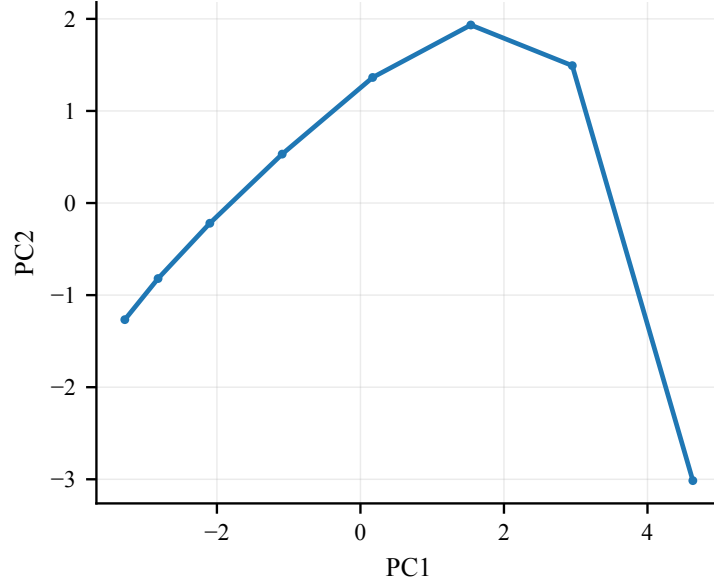


Figure 14: Latent PCA across ACT steps (full TRM representative seed).

E Grokking dynamics and progress measures

Figure 16 shows train/test accuracy and loss across training (delayed generalization), and Figure 17 shows the restricted (trigonometric) and excluded losses tracking the full loss post-grokking, mirroring Figure 1 of Nanda et al. [2023].

F Mechanistic circuit analysis

We analyze W_E , W_U , logit-map DFT energy, W_L neuron-logit maps, MLP neuron clustering (R^2 to $\cos / \sin(\omega_k(a+b))$), and attention routing.

G Multi-seed robustness

We train seeds $\{0, 1, 2, 3, 4\}$ for 50,000 steps; per-seed accuracy, best-FVE, and grokking curves appear in the main text (Figures 3 and 9).

H Combined phase plot (Nanda Fig. 7 style)

I Frequency ablations

1. **Trig (key):** retain only key-frequency components.
2. **Excluded (key):** remove key components sequentially.
3. **All-frequency grid:** retain/ablate each $k \in \{1, \dots, P/2\}$ individually.

The all-frequency ablation grid appears in the main text (Figure 8).

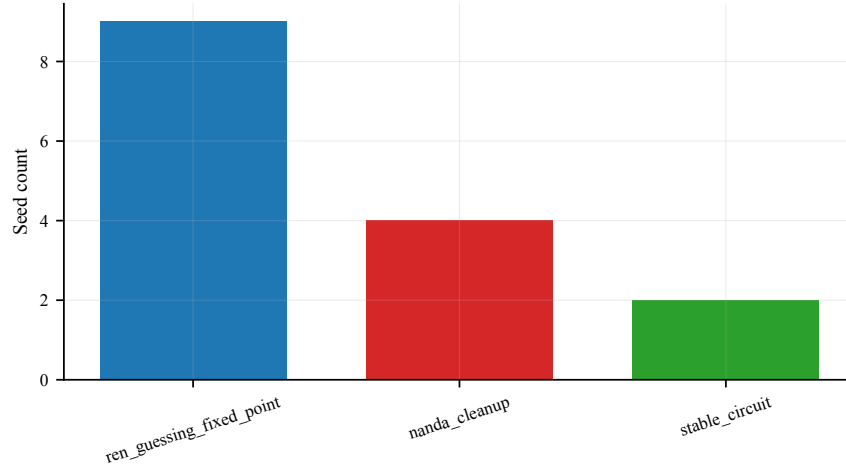


Figure 15: Per-seed post-grokking mechanism classification.

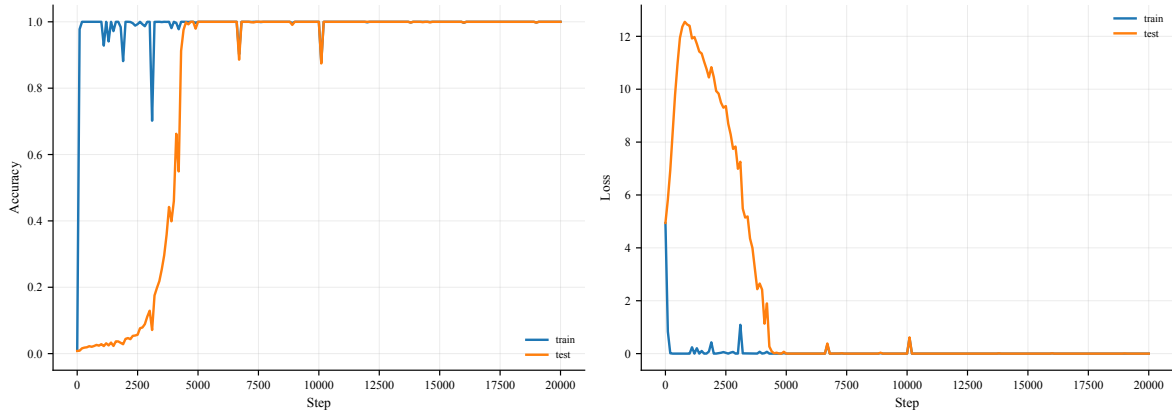


Figure 16: Train/test accuracy and loss vs. step (TRM minimal seed 0).

J Hyperparameters

K HRM / latent recursion probes

Fixed-point violation rate, reasoning-mode counts, mean-field CE by ACT step, and ACT-depth ablations.

L Detailed progress-measure methodology

This appendix mirrors Section 5 of [Nanda et al. \[2023\]](#) with TRM-specific implementation notes.

L.1 Logit grid extraction

For each pair (a, b) in the modular-addition dataset, TRM receives input tokens $[a, b, =]$. We run a single forward pass with ACT halted and read logits at position 2 (the “=” token). Stacking over all P^2 pairs yields $L \in \mathbb{R}^{P \times P \times P}$ where $L_{a,b,c}$ is the logit for output c .

Table 5: Post-grokking mechanism classification (Ren + Nanda probes).

Model	Seed	Primary mechanism	Ren applies?
trm_full_a	seed_0	ren_guessing_fixed_point	Yes
trm_full_a	seed_1	ren_guessing_fixed_point	Yes
trm_full_a	seed_2	ren_guessing_fixed_point	Yes
trm_full_a	seed_3	ren_guessing_fixed_point	Yes
trm_full_a	seed_4	ren_guessing_fixed_point	Yes
trm_full_b	seed_0	nanda_cleanup	Yes
trm_full_b	seed_1	ren_guessing_fixed_point	Yes
trm_full_b	seed_2	ren_guessing_fixed_point	Yes
trm_full_b	seed_3	ren_guessing_fixed_point	Yes
trm_full_b	seed_4	ren_guessing_fixed_point	Yes
trm_minimal	seed_0	stable_circuit	Null
trm_minimal	seed_1	stable_circuit	Null
trm_minimal	seed_2	nanda_cleanup	Null
trm_minimal	seed_3	nanda_cleanup	Null
trm_minimal	seed_4	nanda_cleanup	Null

L.2 Fourier basis on cyclic groups

For prime modulus $P = 113$, define $\omega_k = 2\pi k/P$. Basis functions on index $x \in \{0, \dots, P-1\}$:

$$\phi_k^{\cos}(x) = \cos(\omega_k x), \quad \phi_k^{\sin}(x) = \sin(\omega_k x). \quad (1)$$

For the logit map, the trigonometric identity $\cos(\omega_k(a+b-c))$ decomposes into products of 1D Fourier components along a , b , and c .

L.3 Restricted (trigonometric) loss

Given key frequencies $K = \{k_1, \dots, k_5\}$, we project L onto the span of $\cos(\omega_k(a+b))$ and $\sin(\omega_k(a+b))$ directions for $k \in K$, apply bias correction per [Nanda et al. \[2023\]](#), and compute cross-entropy against labels on train and test splits.

L.4 Excluded loss

For each $k \in K$ in descending sensitivity order, we subtract the rank-1 component associated with frequency k from logits and measure test CE. High excluded loss at step 2000 and low excluded loss at convergence indicate circuit formation.

L.5 Embedding and unembedding Fourier norms

$W_E \in \mathbb{R}^{P \times d}$: apply 1D DFT along the token axis (rows), sum ℓ_2 energy per frequency pair $(k_{\cos}, k_{\sin})$. $W_U \in \mathbb{R}^{P \times d}$: same along the vocabulary axis of the output head. Key frequencies are those with largest excluded-loss sensitivity (not simply largest embedding norm).

L.6 Gini coefficient

Following Nanda Appendix B, we compute the Gini coefficient of the vector of embedding Fourier norms to track sparsification during grokking.

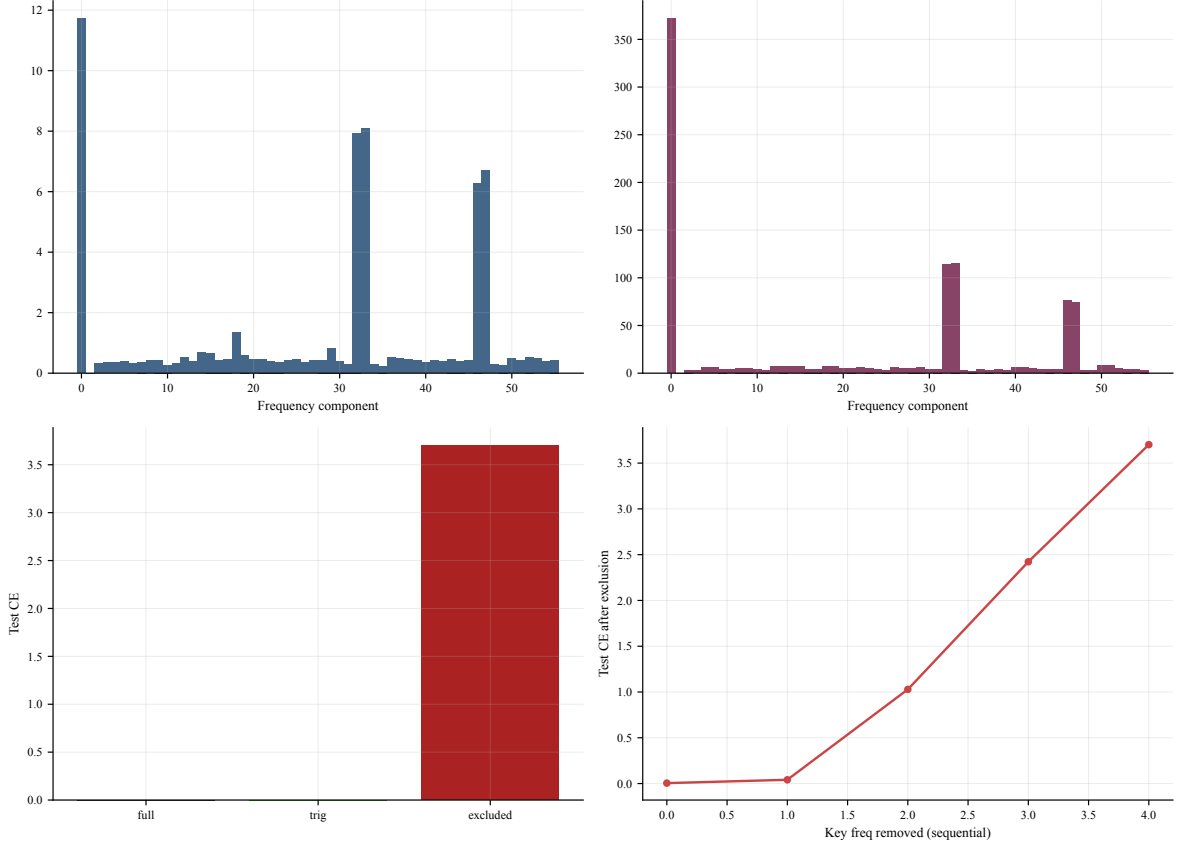


Figure 17: Restricted, excluded, and full cross-entropy vs. step.

M Reverse-engineering methodology

M.1 W_L neuron-logit map

For SwiGLU layer ℓ with down-projection $W_{\text{down}}^{(\ell)} \in \mathbb{R}^{d \times d_{\text{ff}}}$ and unembedding $W_U \in \mathbb{R}^{P \times d}$:

$$W_L^{(\ell)} = (W_{\text{down}}^{(\ell)})^\top W_U^\top \in \mathbb{R}^{d_{\text{ff}} \times P}. \quad (2)$$

Each row of W_L maps one MLP neuron to logits over c ; we DFT each row along the P -axis.

M.2 MLP neuron clustering

For all (a, b) pairs, record post-activation MLP outputs $h_n(a, b)$ at the “=” token. Assign neuron n to frequency k maximizing $R^2(h_n, \cos(\omega_k(a + b)))$ or $\sin(\omega_k(a + b))$.

M.3 Attention decomposition

For layer 0, extract mean attention weights from query position 2 (“=”) to positions 0 (a) and 1 (b), averaged over all (a, b) and batch. Compare per-head routing strength to Nanda’s Figure 4.

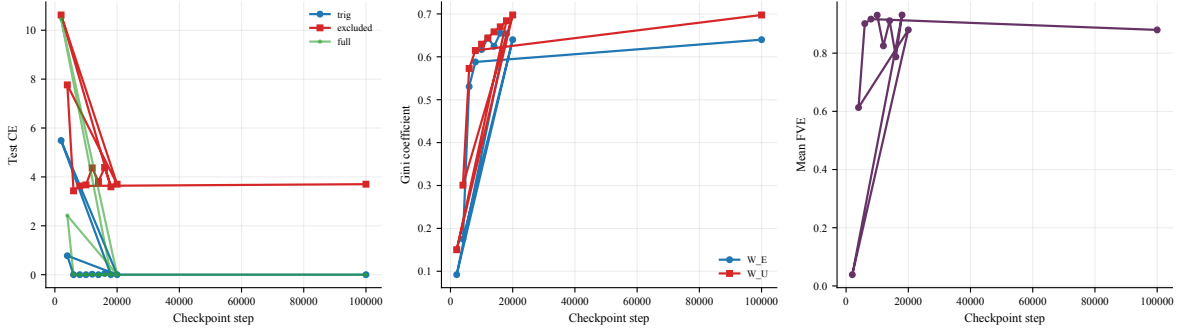


Figure 18: Progress measures across saved checkpoints (TRM minimal seed 0).

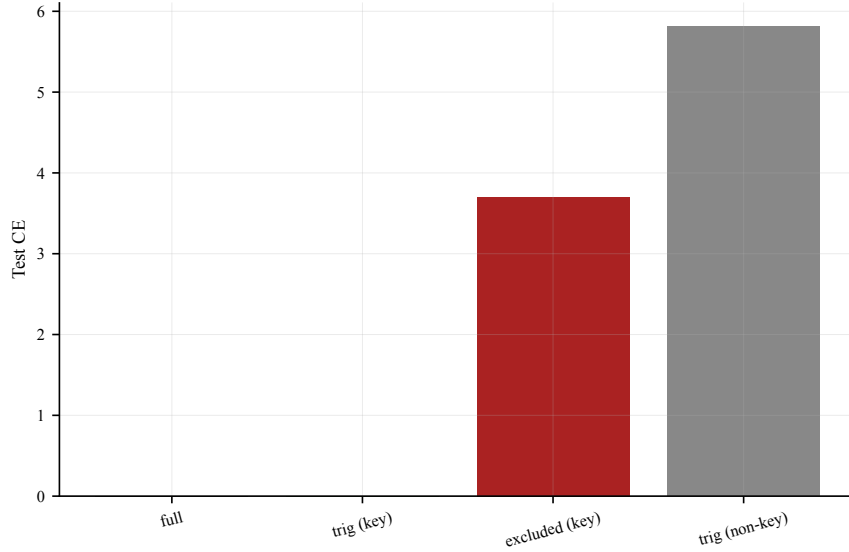


Figure 19: Causal key-frequency ablations at convergence.

M.4 Component patching

We patch attention outputs and MLP outputs separately, measuring change in test CE relative to baseline, following the causal scrubbing spirit of [Nanda et al. \[2023\]](#) Section 4.3.

N Latent probe methodology

N.1 ACT outer loop

TRM uses adaptive computation time with maximum 8 steps. At each ACT step t , we record z_H at the “=” token and the predicted logit distribution.

N.2 Fixed-point violation

For test examples where step 0 prediction is correct, we flag violation if any later step changes the argmax prediction.

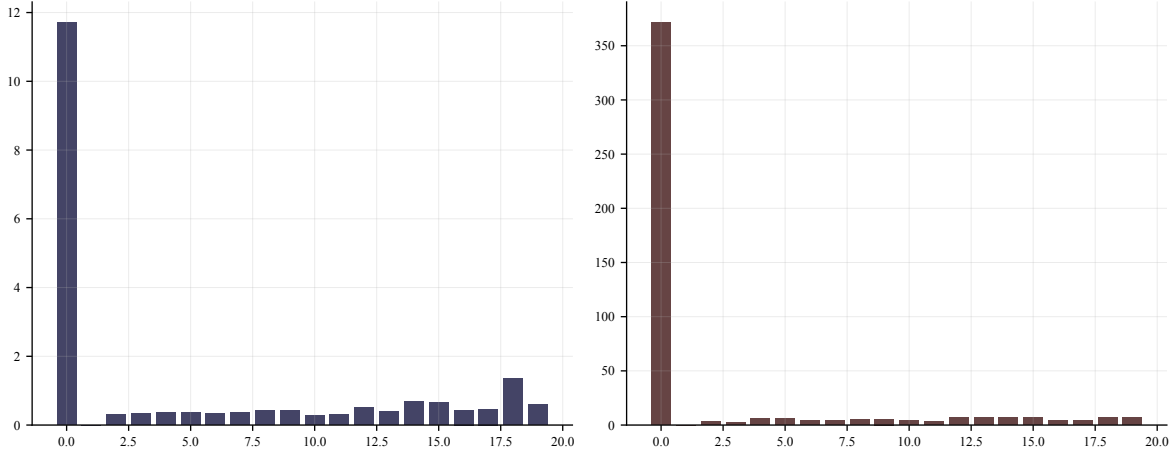


Figure 20: Embedding and unembedding Fourier components.

Table 7: Training hyperparameters (hybrid rigor, 50k steps).

Parameter	Value
P	113
Train fraction	0.3
Optimizer	AdamW
Learning rate	10^{-3}
Weight decay	1.0
β_1, β_2	0.9, 0.98
Steps	50,000
Seeds	5
Hidden size	128
Heads	4
ACT max steps	8 (full TRM)

N.3 Reasoning modes

Following [Ren and Liu \[2026\]](#): **trivial success** (correct at step 0), **nontrivial success** (wrong then correct), **nontrivial failure** (never correct), **oscillation** (alternating predictions).

N.4 PCA trajectories

For individual examples, stack $z_H^{(t)} \in \mathbb{R}^d$ across ACT steps and apply PCA for visualization.

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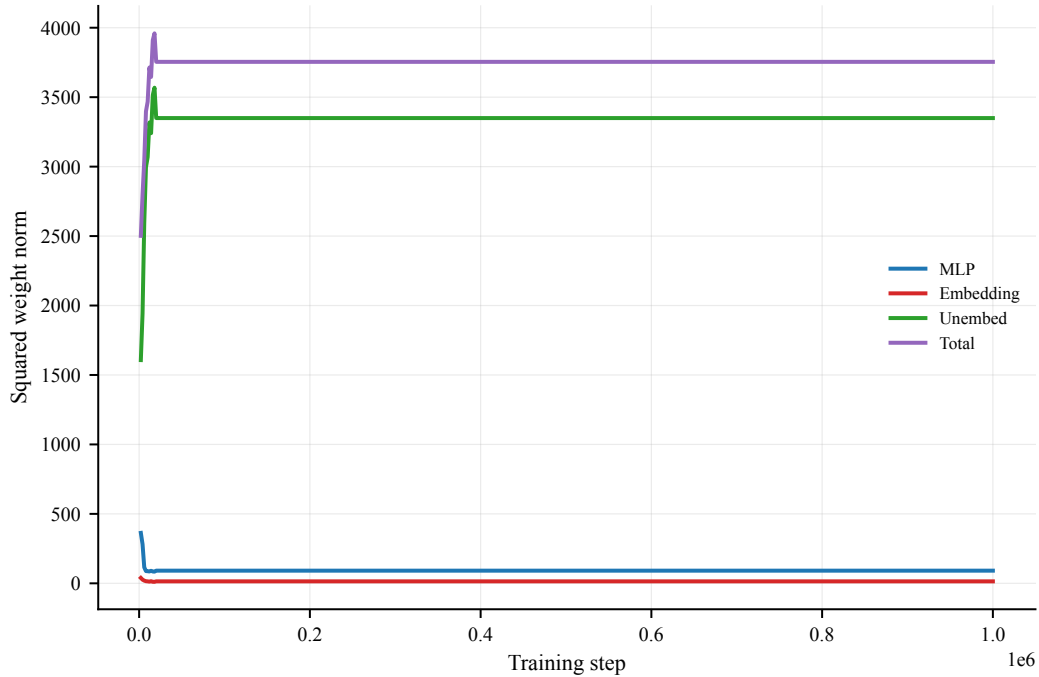


Figure 21: Weight-norm trajectory annotated with memorization, circuit-formation, and cleanup phases (Nanda et al., 2023, Fig. 7 style).

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